Chapter 2

Inventory Management Models

2.1 Introduction

Even with a good grasp of the magnitude and direction of the randomness affecting the future demand, there is always a decision behind predicting the future demand. Allocating resources to match random demand is a difficult problem. The difficulty lies in that resources are often fixed and not flexible: Determining the production capacity of a building, ordering a fixed amount of materials to manufacture products, or scheduling workers ahead of time to service a facility or store. What all these decision have in common is that there is an inherent trade-off between allocating too many resources or allocating too little. On one hand, allocating too many resources is costly and inefficient, and on the other, allocating too little, leads to congestion and lost sales. Both are undesirable outcomes that have to be balanced in an optimal manner. In this chapter, we present strategies to balance optimally these trade-offs in a data-driven way.

2.2 The Newsvendor Model

Following our exposition of characterizing the demand as a Stochastic Process $\{D_t\}_{t=0}^{\infty}$, one of the most popular production models is known as the Newsvendor Model [1]. In this model, a production or inventory decision quantity q has to be taken at time $t = 0$ to satisfy a random demand D_T at horizon time T. The goal of the model is to determine what is the optimal quantity of product that should be produced/ordered. Product is sold at a price p with a unit cost c. The objective function of the seller is their expected profit, defined as:

$$\max_q \mathsf{E}[(D_T \wedge q)p - qc], \tag{2.1}$$

The revenue of the firm is the minimum operator (denoted by $a \wedge b = \min(a, b)$) between the production/inventory quantity q and the random demand D_T multiplied by the price p. As the

production cost of each unit is c, the total variable cost is qc. Subtracting the cost from the revenue gives the profit (note that fixed costs do not affect the production quantity as they do not depend on q).

The first step to solve this problem is to note for any numbers $a, b \in \mathbb{R}$ we have $a \wedge b = a - (a-b)^+$ where $(x)^+ = \max(x, 0)$. With this fact, the profit can be re-written as $(D_T \wedge q)p - qc = [q - (q - D_T)^+]p - qc = q(p-c) - p(q - D_T)^+$. Then, the profit can be re-written as:

$$\max_q q(p-c) - p\mathsf{E}(q - D_T)^+, \tag{2.2}$$

To solve the problem we need to calculate what is the critical point q^* such that the first derivative of the profit is equal to 0. We also need to check that the function is concave, as this will ensure that the critical point of the unique optimal solution maximizing the profit. Taking derivate and equating to 0, yields $\frac{\partial \mathsf{E}(q-D_T)^+}{\partial q} = \frac{p-c}{p}$. We can use our definition of expected value (in Section 1.2) to write $\frac{\partial \mathsf{E}(q-D_T)^+}{\partial q}$ as follows:

$$\frac{\partial}{\partial q}\sum_{i=0}^{\infty}(q-i)^+ p_i = \frac{\partial}{\partial q}\sum_{i=0}^{\infty}(q-i)p_i \mathbf{1}\{i \leq q\} = \frac{\partial}{\partial q}\sum_{i=0}^{q}(q-i)p_i = \mathsf{P}(D_t \leq q) = F_{D_T}(q). \tag{2.3}$$

The first equality is just a re-writting of the event $(x)^+ = x\mathbf{1}\{x \geq 0\}$. The second one just changes the limit of the sumation where the inner term is non-negative. The third equality exploits the fact that the derivative is a linear operator[1] and that the derivative of qp_i is p_i with respect to q. The second derivative is equal to 0, thus making the profit function concave and ensuring that the critical point in indeed the quantity that maximizes the profit. The optimal quantity is:

$$q^* = F_{D_T}^{-1}\left(\frac{p-c}{p}\right), \tag{2.4}$$

$F_{D_T}^{-1}$ denotes the inverse cdf of the distribution of the demand. Connecting this to the demand distribution Section 1, for example by taking the data-driven estimate of the demand as the approximation $D_T = \hat{D}_T(\theta) + V_T \overset{\mathrm{d}}{=} \mu + \sigma Z$ with $\mu = \hat{D}_T(\theta)$ (in general, an arbitrary functional form for the demand. With $\theta = A, \beta$ we can take this as an instance the *base model*) and $\sigma = \sqrt{\mathsf{Var}(V_T)}$. With this approximation, the optimal quantity becomes:

$$q^* = \mu + \Phi^{-1}\left(\frac{p-c}{p}\right)\sigma. \tag{2.5}$$

Where Φ^{-1} is the inverse cdf of the normal distribution. The interpretation of this expression is that the optimal quantity q^* is the predicted value from the model $\mu = \hat{D}_T(\theta)$, plus a buffer proportional to the standard deviation of the demand $\sigma = \sqrt{\mathsf{Var}(V_T)}$. The multiplier $\Phi^{-1}\left(\frac{p-c}{p}\right)$ balances how aggressive the provisioning of inventory should be. First, the price p is normally higher than the cost c, that is, $p > c$. To get a feel for the multiplier, suppose the price is

[1] The derivative of a sum is the sum of the derivatives of its terms.

3 times the cost, or $p = 3c$ (a typical pricing strategy in the absence of information), then $(p-c)/p = 2/3$. Evaluating the inverse at this value gives $\Phi^{-1}(2/3) \approx 43\%$, meaning that the extra buffer above the predicted demand μ is 43% of its standard deviation σ. The larger the difference between the price and cost is $p - c$, the greater the amount of provisioning should be (because the loss of profit from a sale lost is orders of magnitude higher than the cost). As we will see next, we can modify this model to more complex situations where there are costs associated with unsold product. See Figure 2.1 for a visual representation.

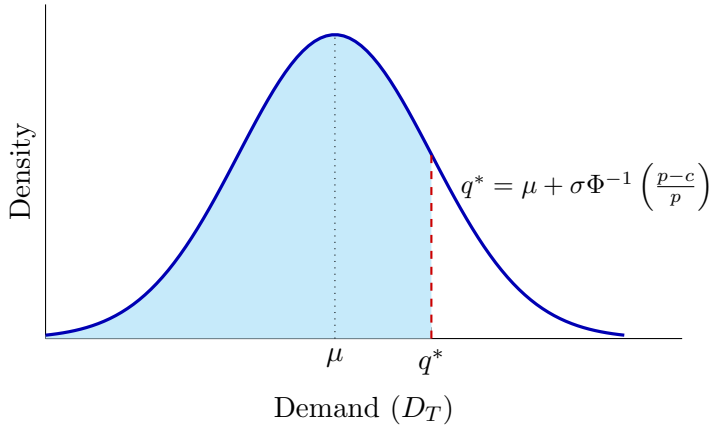

Figure 2.1: Optimal inventory level q^* in the Newsvendor Model. The shaded area represents the probability of meeting all demand, corresponding to the critical ratio $(p-c)/p$.

Example 2.1 (Avocado Toast Restaurant). A national restaurant chain serves a signature avocado toast dish in their menu. The restaurant procures avocados from a wholesale producer and can purchase avocados at a price $c = \$0.68$ per unit. Every avocado dish is sold at a price $p = \$12$, other variable costs add up to $c_a = \$3$ per dish (other ingredients and labor). Every unused avocado has to be disposed at a cost of $c_d = \$0.20$. The distribution of the demand at time T is estimated as a normal random variable with mean $\mu = 70$ and standard deviation $\sigma = 10$ dishes for a specific location of the restaurant. Also, when customers arrive and the avocado toast is sold-out, with probability $p_r = 1\%$ they never come back (imagine a binary random variable B_r with $\mathsf{P}(B_r = 1) = p_r = 1 - \mathsf{P}(B_r = 0)$, independent of the demand). The average lifetime profit per customer is estimated to be $\ell = \$200$. What is the optimal quantity of avocados that maximizes the expected profit for this location? What happens when the lifetime profit per customer increases to $\ell = \$2,000$?

The expected profit for the restaurant can be written as:

$$\max_q \mathsf{E}[(D_T \wedge q)p - q(c + c_a) - (q - D_T)^+ c_d - (D_T - q)^+ B\ell], \tag{2.6}$$

Rearranging terms (noting $a \wedge b = a - (a-b)^+$ as in Equation (2.2)) leads to this expression:

$$\max_q q(p - c - c_a) - (c_d + p)\mathsf{E}(q - D_T)^+ - p_r \ell \mathsf{E}(D_T - q)^+, \tag{2.7}$$

Similar to the previous example, the strategy to solve the problem is to take derivatives with respect to q and equate to 0. The derivative $\frac{\partial \mathsf{E}(D_T - q)^+}{\partial q}$ is equal to $-\bar{F}_{D_T}(q) = F_{D_T}(q) - 1$ (the trick is the same as in Equation (2.3), noting that the expression is non zero when the demand is between q and ∞). Grouping terms leads to the following optimal demand quantity:

$$q^* = F_{D_T}^{-1}\left(\frac{p - c - c_a - p_r \ell}{p + c_d - p_r \ell}\right). \tag{2.8}$$

In the case of this example, this leads to $q^* = \mu + \sigma \Phi^{-1}\left(\frac{p-c-c_a-p_r\ell}{p+c_d-p_r\ell}\right) \approx 76.03$ avocados. When the lifetime profit per customer ℓ is \$2,000 the optimal quantity rises to $q^* \approx 81.72$ avocados.

2.2.1 Empirical Distributions

The Newsvendor model relies heavily on the accuracy of the inverse cdf $F_{D_T}^{-1}$. In real life, the true distribution of the demand is rarely known, and the estimation has to be done on the empirical distribution from a sample. The empirical distribution of the demand is constructed by first sorting observations of the demand in ascending order. A sample of i.i.d. observations of the demand $\{d_i\}_{i=1}^N$ (here, the big assumption is that these are all independently sampled observations of the demand D_T). Let $d_{(i)}$ be the i-th smallest element of the sample, such that $d_{(1)} \leq d_{(2)} \leq \cdots \leq d_{(N)}$. The *Empirical probability measure* assigns each sample a equal probability $1/n$. The sample or empirical mean is then defined as $\frac{1}{n}\sum_{i=1}^n d_i$ which by the LLN converges to the true mean as we discussed in the previous chapter. Likewise, the empirical cdf is defined as $\hat{F}_n(q) = i/n$ for $i = \text{argmax}\{i : d_{(i)} \leq q\}$. Similarly, the empirical inverse cdf is defined as $\hat{F}_n^{-1}(p) = d_{(i)}$ for $i = \text{argmin}\{i : i \geq pn\}$. A natural question is how far off is the Newsvendor inventory quantity when using the empirical cdf in lieu of the real one.

A useful bound for the divergence of the empirical cdf and the actual one was developed in two papers [2, 3] and is commonly known as the *DKW bound* (due to the authors of the first paper, which was later refined by the second one). The bound is stated as follows:

$$\mathsf{P}\left(\sup_q |F(q) - \hat{F}_n(q)| > \lambda\right) \leq 2e^{-2n\lambda^2}, \tag{2.9}$$

What this bound states is that the probability that the deviation from the empirical and the actual cdf exceeds a value λ (for any value q) is less than $2e^{-2n\lambda^2}$. As n increases, this probability decays to 0. Meaning that as the sample size increases, the accuracy of the empirical cdf converges to the true one. Interestingly, we can use this inequality to build confidence intervals on the empirical cdf. For example, with probability $1-\alpha$ the true value of the cdf $F(q)$ is within $\hat{F}_n(q) \pm \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)}$ (hint: equate the right hand side of (2.9) to α) for any q.

Going back to our topic of interest, we want to know how far is the solution of the empirical Newsvendor model $\hat{q}_n^* = F_n^{-1}(\gamma)$ with respect to the real solution $q^* = F^{-1}(\gamma)$ (here, γ denotes the optimal constant in the Newsvendor model, for example $\gamma = \frac{p-c}{c}$ in the plain vanilla formulation). In Section 1 we emphasized that event calculus is a powerful problem solving tool. That is, to write complex events in terms of events that we know well probabilistically. In this case, we know about bounding the empirical cdf in Equation (2.9) and need to find the right event to express the relation to the empirical inverse cdf $F_n^{-1}(\gamma)$.

The optimal quantity $q^* = F^{-1}(\gamma) = \inf\{q : F(q) \geq \gamma\}$. Now we know that $F(q)$ is within $\hat{F}_n(q) \pm \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)}$ with probability at least $1-\alpha$. Putting these two facts together we get that $q^* \in \inf\{q : \hat{F}_n(q) \pm \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)} \geq \gamma\}$ or $q^* \in F_n^{-1}\left(\gamma \pm \sqrt{\frac{1}{2n}\ln\left(\frac{2}{\alpha}\right)}\right)$ with probability $1-\alpha$. Note that this bound is not symmetric and not centered at $F_n^{-1}(\gamma) = q_n^*$. See Figure 2.2.

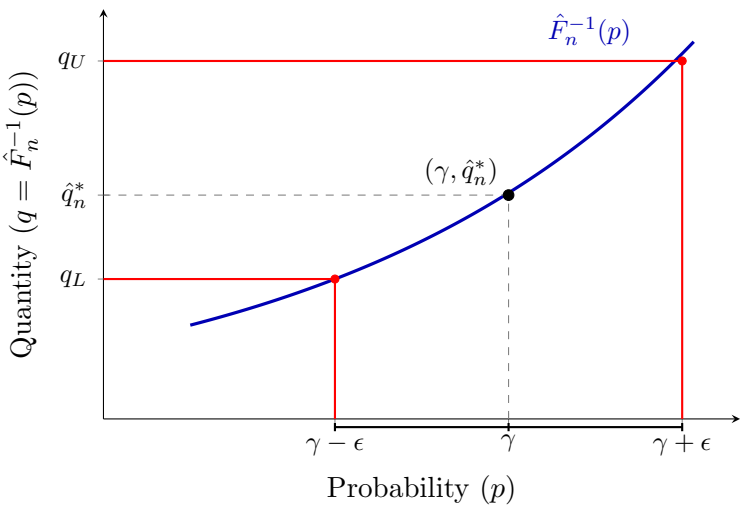

Figure 2.2: Mapping the DKW confidence interval. The symmetric interval $\gamma \pm \epsilon$ results in a non-symmetric confidence interval $[q_L, q_U]$ due to the nonlinearity of the inverse CDF.

Often, it is also a natural question what should be the average profit using a quantity $\hat{q}$ (for example, the optimal sample inventory q_n^*) given a series of n historical observations of the demand. The sample average profit $\mu_\pi(n)$ of the plain vanilla Newsvendor model is:

$$\mu_\pi(n) = \frac{1}{n}\sum_{i=1}^{n} \hat{q}(p-c) - p(\hat{q} - d_i)^+, \tag{2.10}$$

We know that because of the CLT, this mean will converge to the true average profit $\mu_\pi = \hat{q}(p-c) - p\mathsf{E}(\hat{q} - D_T)^+$. The variance of the profit is $\mathsf{Var}(p[(\hat{q} - D_T)^+]) = p^2\mathsf{Var}(\hat{q} - D_T)^+$, for any distribution of the demand D_T, this variance is at most $q^2/4$[2]. Then, we have the following probabilistic relationship between the sample and true profit:

$$\mu_\pi(n) \stackrel{\mathrm{d}}{=} \mu_\pi + \frac{p\sqrt{\mathsf{Var}(\hat{q} - D_T)^+}}{\sqrt{n}} Z \leq \mu_\pi + \frac{p\hat{q}}{2\sqrt{n}} Z, \tag{2.11}$$

[2]This is the so-called Popovicius's variance inequality. Let X be a random variable such that $\mathsf{P}(X \in [a,b]) = 1$, then, $\mathsf{Var}(X) \leq (b-a)^2/4$.

The first equality is the equality in distribution of the CLT. The second is the result of bounding the variance with the inequality above. The result says that for any distribution of the demand, in the i.i.d. case solving the plain vanilla Newsvendor problem converges to the true demand as n grows, and that the noise is bounded by the product of the price p and the quantity $\hat{q}$ divided by a factor of $2\sqrt{n}$. For example, to obtain a confidence interval with probability $1-\alpha$ of the true average profit μ_π by using quantity $\hat{q}$, we have that the error $|\mu_\pi - \mu_\pi(n)| \overset{\mathrm{d}}{=} \frac{p\hat{q}}{2\sqrt{n}}|Z|$. Then, we want to find the value ϵ such that $\mathsf{P}\left(|Z|p\hat{q}/(2\sqrt{n}) > \epsilon\right) = \alpha$, which is the same as $\mathsf{P}(Z > 2\epsilon\sqrt{n}/(p\hat{q})) = \alpha/2$. Applying the inverse tail cdf of the normal distribution, we get solving for ϵ:

$$\mu_\pi \in \mu_\pi(n) \pm \bar{\Phi}^{-1}(\alpha/2)\frac{p\hat{q}}{2\sqrt{n}} \text{ w.p. } 1-\alpha, \tag{2.12}$$

For example, choosing $\alpha = 5\%$, $\bar{\Phi}^{-1}(\alpha/2) = 1.96$, meaning that the true profit is with 95% probability within $1.96\frac{p\hat{q}}{2\sqrt{n}}$ of the sample profit in the plain vanilla Newsvendor model. This technique can also be used for all variations of the Newsvendor problem in the i.i.d. case, noting that the complement variance $\mathsf{Var}(D_T - \hat{q})^+$ is also bounded.

These bounds are very useful because the do not depend on the distribution of the demand and allow to infer the true performance of the model as a function of the sample size. We illustrate this in the following example.

Example 2.2 (Launching a new product). A supermarket chain is planning to release a new bottled orange juice. They perform a pilot of 30 days ($n = 30$), observing an average daily profit of \$1,500. The price of the juice is \$3.49 and they produce per store $\hat{q} = 100$. Assuming the distribution of the demand of juices is i.i.d. in the plain vanilla Newsvendor model. The shelf opportunity cost of the supermarket (the minimum profit that a product makes in that shelf area) is \$1,450 per day. With this data, the supermarket wants to conduct a hypothesis test to decide whether to include the orange juice permanently or continue the pilot. What's the probability that the daily profit is less than \$1,450?

With our analysis we know the true profit μ_π is bounded in distribution to $\mu_\pi(n) + \frac{p\hat{q}}{2\sqrt{n}}Z = \$1,500 + (\$349/2\sqrt{30})Z$. Then $\mathsf{P}(\$1,500 + (\$349/2\sqrt{30})Z < \$1,450)$ simplifies to $\mathsf{P}(Z < -\$50(2\sqrt{30})/\$349)$, which is $\mathsf{P}(Z < -1.56) = \Phi(-1.56) = 5.9\%$. Which means that for a 95% confidence one sided test, the pilot should continue (as the probability is above 5%).

2.3 Inventory Replenishment - The (s, S) rule

In the previous section, we proposed a model for optimizing a production quantity such that profit is maximized in a static setting. The methodology can be extended to optimize multiple

products and locations. As well as optimizing production accross time. In this section, we deal with a common inventory policy taking into account the evolution of the demand over time.

In the previous chapter, we characterized the randomness of the demand as a Stochastic Process $\{D_t\}_{t=0}^{\infty}$. Assuming the distribution of the demand is known, define the inventory level at period t by I_t. The inventory can be defined as the quantity of a commodity or product that is stored to service a random demand. The simplest rule to replenish the inventory is known as the (s, S) rule. This rule works as follows: at the end of period t, if the inventory I_t is below a level s. An order (assumed to fulfill instantaneously) replenishes the inventory to a level S. Then, the inventory I_{t+1} is equal to $S - D_{t+1}$ when $I_t \leq s$, and it is $I_t - D_{t+1}$ otherwise (in the case when $I_t > s$). Clearly, the time between inventory replenishments is random and depends on the distribution of the demand.

Under the (s, S) rule, the inventory is in itself a random variable that depends on the demand. The conditional distribution of the inventory is given by:

$$\mathsf{P}(I_{t+1} = j | I_t = i) = \mathsf{P}(D_{t+1} = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j), \tag{2.13}$$

When the distribution of the demand is i.i.d., that is, $D_t \overset{\mathrm{d}}{=} D$ for all t. We can express the inventory process I_t as a Markov Chain[3] with transition probabilities $p_{ij} = \mathsf{P}(D = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j)$. Note that the inventory can take negative values as the demand can exceed the inventory quantity I_t. Conversely, under the (s, S) rule, the inventory never exceeds S by construction. See Figure 2.3.

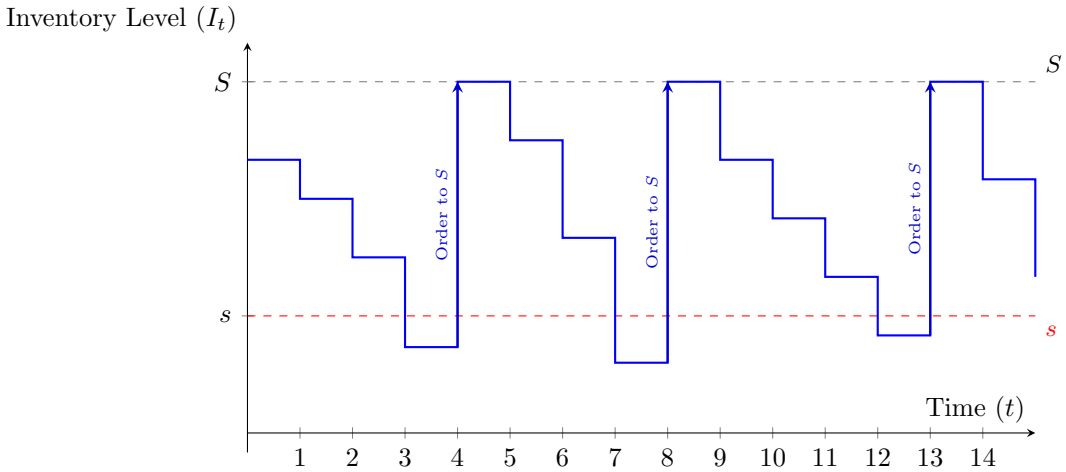

Figure 2.3: Inventory level I_t over time under the (s, S) policy. Replenishment is triggered the moment $I_t < s$, instantly bringing the stock back to the maximum level S.

With these elements we have enough building blocks to express problems of interest: For example, suppose that everytime there is a replenishment of inventory, there is a fixed cost K

[3] A Markov Chain is a special class of Stochastic Processes, where the distribution of the process X_t depends only on the realization of the process the previous period X_{t-1}.

plus a variable cost of c per unit. Likewise, suppose there is a holding cost of c_h per day the inventory is stored. Finally, when the inventory is negative (meaning that there was not enough inventory to service the demand), those sales are lost, a loss of a net revenue $\tilde{p}$ per unit. We can summarize these costs as a function $c(i)$ for each possible state of the inventory:

$$c(i) = \begin{cases} ic_h, & \text{if } s \leq i \leq S, \\ ic_h + K + (S-i)c, & \text{if } 0 \leq i < s, \\ K + Sc - i\tilde{p}, & \text{if } i < 0. \end{cases} \tag{2.14}$$

A Markov Chain transitions between states forever (even when it gets absorbed into a single state). When the transitions can be expressed with a fixed matrix P with elements p_{ij} denoting the probability that the chain at state i transitions to state j. The so-called limiting distribution is given by a vector π as:

$$\lim_{t\to\infty} P^t = \pi \quad \text{or} \quad \pi P = \pi, \tag{2.15}$$

Verbally, the vector π denotes the distribution of the states long-term. For example, $\pi(i) = \lim_{t\to\infty} \sum_{s=1}^{t} \mathsf{E}\mathbf{1}\{X_s = i\}/t$ shows what proportion of time (out of t) is spent in state i for a Markov Chain $\{X_t\}$.

In our case, this limiting distribution π is useful because we know we can express our inventory model as a Markov Chain with transition probabilities $p_{ij} = \mathsf{P}(D = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j)$. Then, the long-term cost of the (s, S) inventory rule is given by:

$$\sum_{i=s-M}^{S} \pi(i)c(i), \tag{2.16}$$

Where M is given by a value such that $\mathsf{P}(D \leq M) = 1$. That is, the largest the demand can be (why is the lower limit of this Markov Chain $s - M$?). To compute the limiting distribution we just need to solve the system of equations $\pi P = \pi$ numerically. We spend the rest of this section discussing how to find actual values for (s, S). See Figure 2.4 for an example analysis.

2.3.1 Finding the (s, S) values in the i.i.d. case

The most direct approach to find a pair of values $0 \leq s < S$ that is optimal in the sense of minimizing is to enumerate all possible combinations of values of s, S and calculate the stationary distribution by solving $\pi P(s, S) = \pi$, where $P(s, S)$ is the transition matrix under the policy (s, S) (recall, the transition probabilities depend on both s and S, as $p_{ij}(s, S) = \mathsf{P}(D = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j)$). We summarize the procedure in the following algorithm:

Algorithm 2 *Finding Optimal* (s^*, S^*) *Inventory Policy - i.i.d. case*

Input: A distribution of the demand $\mathsf{P}(D = i)$. A cost function $c_\theta(i)$ such as the one in Equation (2.14) with parameters θ. And upper limit of inventory capacity S_{max}, and for the demand M.

Output: The optimal inventory policy parameters (s^*, S^*).

procedure OPTIMALINVENTORY(F_D, c_θ, S_{max})

  $S, s^*, S^* = S_{max}$

  $c^*, \bar{c} = \infty$

  for $S = 1$ **to** S_{max} **do**

    for $s = 0$ **to** $S - 1$ **do**

      $P \leftarrow [p_{ij}] = [\mathsf{P}(D = S\mathbf{1}\{i < s\} + i\mathbf{1}\{s \leq i \leq S\} - j)]$ for $i, j \in \{s - M, \cdots, S\}$.

      Solve π such that $\pi = \pi P$ and $|\pi| = 1$.

      Update and compute $\bar{c} \leftarrow c'_\theta \pi$.

      if $\bar{c} < c^*$ **then**

        Update $c^* \leftarrow \bar{c}, s^* \leftarrow s, S^* \leftarrow S$.

      end if

    end for

  end for

  return s^*, S^*, c^*.

end procedure

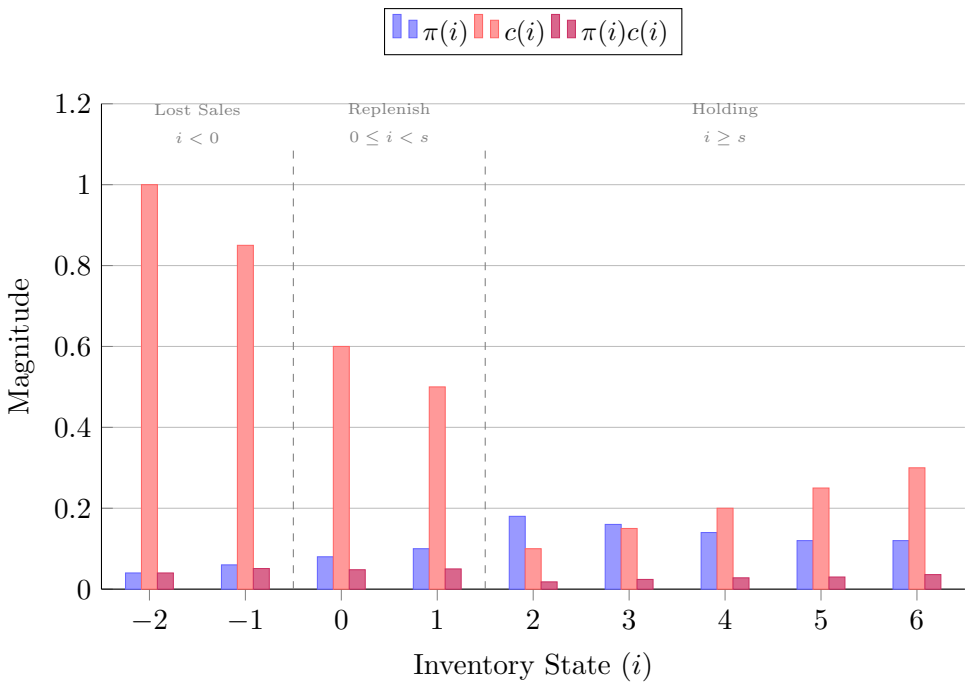

Figure 2.4: Long-term inventory cost analysis. The purple bars represent the weighted cost contribution of each state to the total expected cost $\sum \pi(i)c(i)$. Optimization aims to choose s and S such that the high-probability states (π) align with low-cost states (c).

Given the structure of the transitions of the inventory, this is not the most efficient algorithm to solve the inventory policy, but it is the easiest to implement as a computational routine. Next we examine the case to find policies for time-varying demands.

2.3.2 The time-varying demand optimal inventory

The key observation in the previous subsection was that the inventory process is a Markov Chain, that is, that the transition probabilities are time-homogeneous. As we explored in Section 1, demand is often seasonal and time-dependent. In this section, we present a useful technique to solve problems that have a time-dependent structure.

In the time-dependent setting, the demand distribution $F_{D_t}(i) = \mathsf{P}(D_t \leq i)$ changes from period to period. Suppose we want to derive an inventory policy, that at each period tells us the optimal ordering level that minimizes some cost (for example, the cost function in Equation (2.14)). Given a time horizon of T periods in the future. The objective is to minimize a cost over the time horizon as:

$$\mathsf{E}\left[\sum_{t=1}^{T} c_t(I_t)\right], \tag{2.17}$$

The trajectory of the inventory I_t depends on the previous periods and is not time homogeneous, making the problem seemingly hard to solve. A useful technique to solve sequential

problems is to look one step ahead (or behind) and try to define a recursive relation that allows to solve the problem going to the last period T (where the decision is normally trivial) and solve the problem *backwards*, in a recursive way. The easiest way to set up such relation is to define the *action* that can be taken separated from the *state* of the system, so that at each step the problem reduces to find the optimal action at that state in that time step. The state of the system in this case is already defined as the inventory level I_t, an integer (potentially negative) denoting the inventory amount at the end of period t. In the previous section, for the (s, S) policy, the action was coded in an indicator function (as $I_{t+1} = S\mathbf{1}\{I_t < s\} + I_t(1 - \mathbf{1}\{I_t < s\}) - D_{t+1}$ or $I_{t+1} = I_t + (S - I_t)\mathbf{1}\{I_t < s\} - D_{t+1}$). Inspired by this, we can express the state and the action of the system $I_{t+1} = j - D_{t+1}$ such that $j \geq I_t = i$. In this case, j represents the new level of replenishment of the inventory (or rather, the inventory at the beginning of $t+1$, before the demand D_{t+1} is subtracted). The strategy is then to write a function $\bar{C}_t(i)$ that represents the optimal expected cost from time t to T, that is, $\bar{C}_t(i) = \mathsf{E}\left[\sum_{s=t}^{T} c_s^*(I_s)|I_t = i\right]$, where c^* is the optimal cost of taking the best inventory actions. The strategy is then, to write down a one-step recursion that allows us to break down the problem to a single replenishment decision at every time t. With our representation, we can express the optimal cost between states and actions by the following relation:

$$\bar{C}_t(i) = \min_{j \geq i} \mathsf{E}\left[c(i, j) + \bar{C}_{t+1}(j - D_{t+1})\right], \tag{2.18}$$

The interpretation of this equation is that the optimal cost $\bar{C}_t(i)$ at time t when the last inventory $I_t = i$ is the expected cost of that state i and the optimal action j, for that period in the function $c(i, j)$. Plus, the expected optimal cost of the new inventory at a level $j - D_{t+1}$. The minimum denotes choosing the action $j \geq i$ that minimizes this cost.

With this recursion, it is possible to solve for all $C_t(i)$ for all $i = -M, \ldots, S_{max}$ and times $t = 0, 1, \ldots, T$, starting from the end at time T. And populating entries backwards (it is easier to imagine a matrix $\bar{C}$ with entries $c_{i,t} = \bar{C}_t(i)$). This problem solving technique is known as *Dynamic Programming* and Equation (2.18) is a *Bellman Equation*[4]. Storing the optimal replenishment levels $j^*_{t,i} = \text{argmin}_{j \geq i} \mathsf{E}\left[c(i, j) + \bar{C}_{t+1}(j - D_{t+1})\right]$ is also a good idea, as they will be a *dictionary* to the optimal replenishment level when the inventory is i at time t. See Figure 2.5 to visualize the dynamic inventory policy.

The modeling aspect is captured in the cost function $c(i, j)$. The goal is to modify this function to correctly capture the underlying business problem that we want to solve. We illustrate this with the following example:

Example 2.3 (Oxygen Concentrator Distribution Center). An oxygen concentrator company that delivers oxygen concentrators (a device used for patients that need a constant flow of oxygen while they are at home) rents a warehouse to be a distribution center for the concentrators in a major city. They have a random seasonal daily demand for concentrators D_t,

[4] A equation representing a recursive relation of the optimal decision of a dynamic problem.

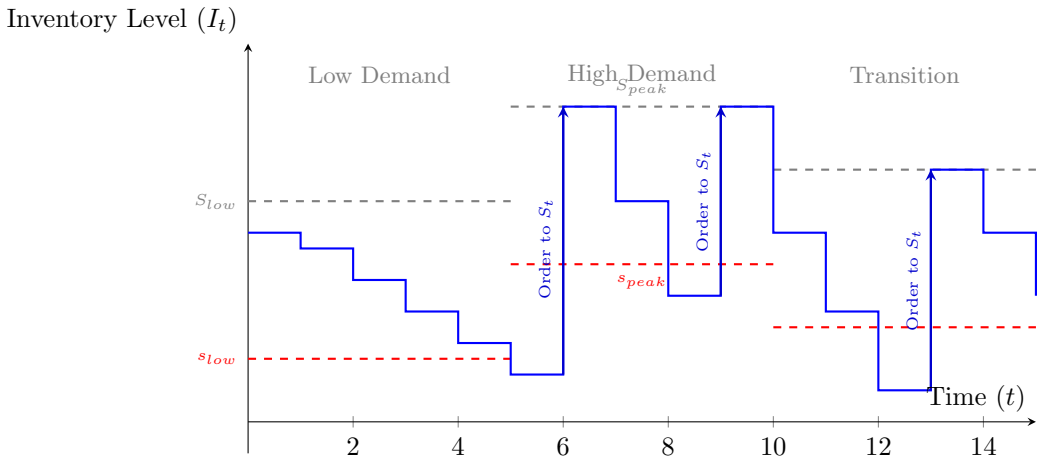

Figure 2.5: Inventory process with time-varying demand. As the depletion rate (slope) increases during peak periods, the Dynamic Programming solution adjusts the optimal s_t and S_t levels to minimize expected costs over the horizon T.

that among other features, depends on the atmospheric humidity level X_t (there is a correlation between respiratory diseases and the level of humidity in the air). The cost structure of running the warehouse works as follows: orders that are fulfilled with stock from the warehouse have a unit cost of c dollars (using an in-house delivery system). Orders that are fulfilled when the demand exceeds the stock cost $\tilde{c}$ dollars using an external courier that picks up a concentrator in another facility and delivers it to the client, this cost is higher than the in-house system, that is $\tilde{c} > c$. Stock replenishment orders have a fixed cost K per truck plus a unit cost c_r (the lead time is overnight, and each truck can store up to L concentrators). Holding the concentrators in the facility has an associated daily cost c_h (including maintenance and the internal logistics of the warehouse that are proportional to the number of concentrators). How to model and minimize this cost over the next month ($T = 30$, assuming today is $t = 0$) assuming a known high-fidelity forecast of the humidity levels X_t for next month $\{x_t\}_{t=0}^{T}$?

The first step is to write the cost function $c(i, j)$ to reflect these costs. For example, $c(i, j) = i^+ c_h + \tilde{c}(D_{t+1} - j)^+ + \mathbf{1}\{j > i\}(\lceil (j^+ - i^+)/L \rceil K + (j^+ - i^+)c_r)$, note the $i^+ = \max(i, 0)$ reflects that when the inventory ends up negative, these orders are not *back-ordered* as they still end up fulfilled at the higher cost $\tilde{c}$. $(j^+ - i^+)$ denotes the order quantity. The quantity $\lceil (j^+ - i^+)/L \rceil$ denotes the number of trucks needed ($\lceil a \rceil$ is the closest integer above a, typically referred in programing as the ceiling function[5]).

The second step is to incorporate the humidity forecast $\{x_t\}_{t=0}^{T}$. To do this, recall in Section 1 that a way to incorporate side-information is to analyze the conditional random variable, that

[5] A definition is $\lceil a \rceil = \min\{x \in \mathbb{Z} : x \geq a\}$.

is, $D_t|X_t$. Calibrating this conditional distribution results in conditional probabilities $\mathsf{P}(D_t = i|\{X_t = x_t\}_{t=0}^T)$. Solving the problem with this side information amounts to calculate the expectations with these conditional probabilities (instead of the unconditional ones). Replacing both the new cost $c(i,j)$ and taking expectations conditional to $X_t = x_t$ in (2.18). Explicitly, this yields the recursion for $\bar{C}_t(i) = \min_{j \geq i} \mathsf{E}[i^+ c_h + \tilde{c}(D_{t+1} - j)^+ + \mathbf{1}\{j > i\}(\lceil (j^+ - i^+)/L \rceil K + (j^+ - i^+)c_r) + \bar{C}_{t+1}(j - D_{t+1})|\{X_t = x_t\}_{t=0}^T]$.